

SOMJIT ROY

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RESEARCH INTERESTS AND SKILL SET

My research advances next-generation statistical modeling frameworks that bridge data-driven learning with scientific reasoning. I develop novel methodologies for complex real-world problems in materials discovery, geophysics, bioinformatics, and finance. Working at the intersection of scientific machine learning and Bayesian modeling & computation, I focus on scientifically guided probabilistic inference, physics-informed modeling, approximate Bayes, tree-based models, and Bayesian optimization, with the goal of building interpretable, scalable, and scientifically grounded learning systems.

Core methods: PROBABILISTIC SciML SYMBOLIC REGRESSION APPROXIMATE BAYES BAYESIAN OPTIMIZATION

EDUCATION AND TRAINING

- 🏛️ **Texas A&M University, College Station, TX, USA** 2023–present
Doctor of Philosophy (Ph.D.) in Statistics GPA: 4.00/4.00
Advisors: **Bani K. Mallick** (TAMU Statistics) & **DebdEEP Pati** (UW–Madison Statistics)
Area of Study: Probabilistic Scientific Machine Learning and Computational Bayes
- 🏛️ **University of Calcutta, Kolkata, West Bengal, India** 2021–2023
Master of Science (M.Sc.) in Statistics, Ranked First Class First GPA: 8.15/10.00
- 🏛️ **St. Xavier's College (Autonomous), Kolkata, West Bengal, India** 2018–2021
Bachelor of Science (B.Sc.) in Statistics, Honours GPA: 8.63/10.00

EXPERIENCE

- 🏢 **Upstart Network, Inc.** June 2026–Aug 2026
Research Scientist Intern Remote
- **Selection-Bias-Aware NPV RMSE for Applicant-Level Underwriting Evaluation.**
 - Advised by **Eric Oden** in the Machine Learning team. Development of a weighted Net Present Value (NPV) RMSE metric to correct selection bias in funded-only underwriting evaluation.
- 🏢 **Los Alamos National Laboratory** May 2025–Aug 2025
Graduate Summer Intern Los Alamos, NM, USA
- **Scalable Physics-Informed Neural Operator for Seismic Wave Modeling.**
 - Advised by **Kai Gao** & **Ting Chen** in the Earth and Environmental Sciences (EES-16) division. Developed a scalable framework for seismic wave modeling using physics-informed neural operators (PINO) to solve the acoustic wave partial differential equation in complex, heterogeneous media.
- 🏢 **Tata Electronics Pvt. Ltd.** May 2021–Aug 2021
Data Science Intern Bengaluru, KA, India
- **Statistical Analysis and Optimization of Sandblasting & Anodizing.**
 - Supervised by **Nagasubramanian Kothandaraman** (Tata Electronics Pvt. Ltd.) & **Subhamoy Maitra** (ISI Kolkata). Identified factors influencing material surface properties, including gloss and texture, and optimized process parameters with machine learning architectures, reducing production time by two man-days.

PAPERS AND MANUSCRIPTS

Tags: ★ Probabilistic SciML ★ Symbolic Regression ★ Variational Inference ★ Bayesian Optimization ★ Quantum Cryptography

6. Roy, S., Dey, P., & Mallick, B. K. (2026). VaSST: Variational Inference for Symbolic Regression using Soft Symbolic Trees. (Accepted at UAI 2026). arXiv ↗ ★★☆☆

5. Roy, S., Jaiswal, P., Bhattacharya, A., Pati, D., & Mallick, B. K. (2026+). On Frequentist Regret Analysis of Fractional Gaussian Process Thompson Sampling. (*Submitted*). [arXiv](#) ↗ ☆
4. Roy, S., Dey, P., Pati, D., & Mallick, B. K. (2025+). Hierarchical Bayesian Operator-Induced Symbolic Regression Trees for Structural Learning of Scientific Expressions. (*Submitted*). [arXiv](#) ↗ [2026 IMS Hannan Graduate Student Travel Award; 2nd place at Best of Statistical Science (BOSS) 2026 Conference (TidBit session); 2026 ASA SBSS Student Paper Award; selected for refereed oral presentation at ASA's SDSS 2026; featured by the Texas A&M College of Arts and Sciences ([news](#) ↗)]. ★ ☆
3. Roy, S., Dey, P., Pati, D., & Mallick, B. K. (2025+). A Generalized Tangent Approximation based Variational Inference Framework for Strongly Super-Gaussian Likelihoods. (*Revision submitted: JASA Theory & Methods*). [arXiv](#) ↗ ☆
2. Kumar, A., Maitra, S., & Roy, S. (2024). Almost Perfect Mutually Unbiased Bases that are Sparse. *Journal of Statistical Theory and Practice*, **18**, 61. [doi](#) ↗ ☆
1. Chaudhury, S., Kumar, A., Maitra, S., Roy, S., & Sen Gupta, S. (2022). A Heuristic Framework to Search for Approximate Mutually Unbiased Bases. In *Cyber Security, Cryptology, and Machine Learning* (CSCML 2022), Lecture Notes in Computer Science, vol. **13301**. Springer, Cham. [doi](#) ↗ ☆

OPEN-SOURCE SOFTWARE

🔗 **GitHub** [VaSST](#) (Roy, S. & Dey, P., 2025); [HierBOSSS](#) (Roy, S. & Dey, P., 2025); [TAVIE-SSG](#) (Roy, S. & Dey, P., 2025)

📦 **CRAN** [bayesestdft](#) (Roy, S. & Lee, S. Y., 2025); [GoodFitSBM](#) (Ghosh, S., Roy, S., & Pati, D., 2024)

TECHNICAL SKILLS

Programming Languages Python, R, Java, C, C++

Statistical & ML Tools RStan, mcmc, coda, PyMC, PyTorch Lightning, TensorFlow, SciPy, scikit-learn

High-Performance Computing Linux/Unix, MPI/OpenMP, CUDA-GPU, PyTorch DDP

Others Git, Markdown, L^AT_EX

AWARDS AND FELLOWSHIPS

2026 **IMS Hannan Graduate Student Travel Award**, Institute of Mathematical Statistics

2026 **SBSS Student Paper Award**, American Statistical Association

2026 **SDSS Student and Early Career Scholarship**, American Statistical Association

2026 **Best of Statistical Science (BOSS) TidBit Award**, Texas A&M University

2026 **Graduate Research and Presentation (RAP) Travel Award**, Texas A&M University

2024 **NSF Travel Grant**, for presenting at IISA 2024

2024 **Targeted Proposal Teams (TPT) Grant**, Research Assistantship | Texas A&M University

2024 **Selected for CMS³-FAST Summer School**, Texas A&M University

2022 **R. C. Bose Memorial Book Prize**, Calcutta Statistical Association

2022 **IASc-INSa-NASI Science Academies Summer Research Fellowship**, Indian Academy of Sciences

2022 **OPHI Summer School Grant**, University of Oxford

2022 **IAOS Conference and Travel Grant**, World Bank

TALKS AND PRESENTATIONS

2026 **Joint Statistical Meetings** INVITED SBSS AWARD PRESENTATION
Hierarchical Bayesian Operator-Induced Symbolic Regression Trees for Structural Learning of Scientific Expressions Boston, MA

2026 **Symposium on Data Science & Statistics** REFEREED INVITED TALK
Hierarchical Bayesian Operator-Induced Symbolic Regression Trees for Structural Learning of Scientific Expressions Milwaukee, WI

2025 **STAT CAFÉ** SEMINAR TALK
Bayesian Symbolic Trees for Structural Learning of Scientific Expressions College Station, TX

2025 **SIAM Texas-Louisiana Sectional Meeting** INVITED TALK
Physics-Informed Neural Operators for Seismic Wave Modeling Austin, TX

2025 **Los Alamos National Laboratory Student Symposium** POSTER
Physics-Informed Neural Operators for Seismic Wave Modeling Los Alamos, NM

LEADERSHIP AND PROFESSIONAL ACTIVITIES

- **Workflow Workshop Organizer**

Statistics Graduate Student Association, Texas A&M University

Led and coordinated graduate workshops on research workflow, computational practice, and academic productivity.

2024–2025
- **Organizing Committee Member**

Department of Statistics, St. Xavier's College (Autonomous), Kolkata

Served on the organizing committee for the annual departmental fest **epsilon delta**, supporting event planning and coordination.

2019–2020

REFERENCES

Bani K. Mallick
 Department Head & Regents Professor
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 Texas A&M University,
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Debdeep Pati
 Professor
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